A Tensor-Based Hyperspectral Anomaly Detection Method Under Prior Physical Constraints

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***Abstract*— Efficient and precise modeling of the background to accurately identify anomalies is the cornerstone to hyperspectral anomaly detection. Hyperspectral image (HSI) can be regarded as 3-D cube data, which contains both spatial and spectral information. The data are converted into 2-D matrices for processing in most existing method, which loses a large amount of structural information. In addition, it is difficult to construct a model with strong representation ability without enough prior knowledge constraints, and the modeling of the background is easily polluted by anomalies. This article introduces a tensor-based hyperspectral anomaly detection method that takes into account prior physical constraints as a solution to the aforementioned problems. The proposed method uses a tensor representation of the image, which preserves its geometrical properties and adheres to fundamental physical principles. After separating them from the image tensor, the background and anomaly tensors are treated separately. For the background tensor, we introduce segmented smoothness constraints in both spatial and spectral dimensions by applying linear total variation (TV) norm regularization. This can improve resistance to complicated backgrounds and lessen the introduction of extra noise when the background is restored. A low-rank constraint based on image eigenvalues is intended to generate a more realistic background model in its spatial dimension, making the method more sensitive to tiny anomalies. For the anomaly tensor, there is sparsity in its spectral dimension, and the background is effectively separated from the anomaly by l_1-norm constraints. Eventually, the anomaly detection map is decided by the anomaly tensor computed iteratively. Comprehensive experiments on several genuine and simulated datasets show that the proposed method performs significantly better at anomaly detection than the state-of-the-art methods.**

***Index Terms*— Anomaly detection, hyperspectral image (HSI), remote sensing, tensor framework.**

I. Introduction

HYPERSPECTRAL anomaly detection holds a prominent position as a fundamental and significant research focus within the domain of remote sensing image analysis and processing [1], [2], [3], [4]. It can locate potentially suspicious targets in the absence of background and spectral information of anomalies. It is widely used in the fields of disaster warning, precision agriculture, military survey, etc., which is of great significance to economic development and national defense construction [5], [6], [7], [8].

There is no uniform definition of anomalies in hyperspectral images (HSIs) [9], [10]. A target whose spectrum is significantly different from the background spectrum is usually considered as an anomaly. Anomalies are correspondingly divided into local anomalies and global anomalies due to the different areas of the background used for comparison. Low probability of occurrence, fragmented distribution, and significant spectra are the characteristics of anomalies, which are used to distinguish and detect anomalies.

Deep learning has demonstrated outstanding performance in recent years when it comes to extracting features from high-dimensional complicated data. As we are all aware, high-dimensional complex data also include the HSI. As a result, researchers are attempting to use deep learning to the identification of hyperspectral anomalies [11], [12], [13], [14]. These methods fall into two main categories: methods based on reconstruction error [15], [16], [17] and hybrid methods based on feature extraction [18], [19], [20]. For example, the autonomous anomaly detection (Auto-AD) network [17] simplifies the processing by automatically reconstructing the background using a convolutional autoencoder with skip connections, while low-rank embedded network (LREN) [18] develops a trainable density estimation module based on the Gaussian mixture model in the deep latent space to extract image features for constructing a dictionary that characterizes complex HSIs. However, they struggle to address this challenge adequately due to the difficulty in obtaining sufficient training datasets with unknown anomaly samples. The available technical solutions fall short in resolving this issue. Therefore, we concentrate on the traditional anomaly detection method in this article.

According to the different techniques used, the traditional anomaly detection methods can be divided into two types: methods based on statistical models and methods based on representational models. Methods based on statistical models operate under the assumption that the background data adhere to a particular distribution, thereby identifying samples that deviate from this distribution as anomalies. The most classical method is the Reed–Xiaoli (RX) method [21], which was proposed by Reed and Yu in 1990. Also, this method is the benchmark for comparison methods. The method makes an assumption of multivariate Gaussian distribution for the

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background data and detects anomalies by calculating and comparing the Mahalanobis distance of the sample spectrum with the overall background spectrum. The method is simple to calculate and has a wide range of applications. Since the distribution assumption of the background cannot accurately describe the real scene, Kwon et al. [22] proposed the local RX (KRX) method, which obtains local background samples by the double-window technique and calculates the distance between the samples and their spectrum. It further satisfies the distribution assumption by narrowing the range of the background to be compared [23]. Due to the direct use of the original spectrum, the RX method suffers from noise sensitivity. Tao et al. [24] used fractional Fourier transform (FrFT) to preprocess the original image and detect anomalies in the signal of the intermediate domain containing the reflected spectrum and its Fourier domain information. Building on this premise, Ma et al. devised a methodology by combining FrFT with low-rank sparse matrix decomposition (LSMD) to enhance background purification and improve the detection effect [25]. However, these methods do not fundamentally address the problem of real scenes that cannot be accurately described by the distribution assumption.

Methods based on representation models operate under the assumption that background samples can be accurately represented using correlated samples, whereas anomalies cannot be effectively represented in the same manner. The advantage of this type of method over the former is that it does not require background distribution assumptions. In 2015, Li and Du [26] proposed a classical collaborative representation detection (CRD) method that also uses a two-window strategy. It assumes that background samples can be represented by the surrounding background spectrum, while anomalies cannot. This method is more sensitive to the purity of the background spectra. Tu et al. [27] proposed the spatial density background purification (SDBP) method to improve it by removing low-density anomalies and retaining high-density background samples through density peak clustering in 2019. Tan et al. [28] incorporated the inverse distance weight and the Tikhonov regularization into the collaborative representation to purify the background. Hou et al. [29] proposed the CRD with background purification and saliency weight (CRDBPSW) method that uses the least-squares method to solve the initial coefficients and weakens the influence of anomalies on the background. But this type of method makes it difficult to set the size of the double-window without prior knowledge. To address this problem, the researchers incorporated the low-rank representation (LRR) constraint into the model [30], [31], [32]. These methods learn and construct the background dictionary, reconstruct the background image, and then use the residual image to amplify the differences between the background and the anomaly. However, the number of dictionary atoms needs to be set manually and the detection accuracy is very sensitive to the quality of the dictionary.

The LSMD, which seeks to recover low-rank sparse components of observation data under reasonable assumptions, has drawn more attention lately [30], [33], [34]. In hyperspectral anomaly detection, the background is usually composed of a limited number of ground features and is widely distributed, thus making it low-rank in nature. The anomalies satisfy the sparse nature due to their low probability and sporadic distribution. This allows us to separate them using LSMD depending on their different properties. Many published methods based on LSMD can be roughly divided into the following three types according to different low-rank decomposition models: the methods based on robust principal component analysis (RPCA) [35], [36], [37], the methods based on go decomposition (GoDec) [33], [38], [39], and the methods based on LRR [40], [41], [42]. Without sufficient prior knowledge constraints, this type of method is difficult to construct a model with strong representation ability, and the modeling of the background is easily polluted by anomalies.

It is widely known that hyperspectral data can be considered as 3-D cubic data, containing both spatial and spectral information. However, most existing methods transform the data into a 2-D matrix for subsequent processing, which loses the structural detail of the image. Note that the physical quantities described by the tensor remain constant under the change in coordinates. This indicates that the geometric properties and physical principles of the hyperspectral data can be preserved when described with the tensor. Using tensors to improve anomaly detection has become an important research direction. The existing methods can be roughly categorized into the following: methods based on Tucker decomposition [43], [44], [45], methods based on the tensor singular value decomposition [35], [46], and methods based on CANDECOMP/PARAFAC decomposition [37], [47]. However, most of the methods ignore the priori knowledge with physical significance that exists in the image itself, which makes the methods do not have enough constraints to construct accurate background models and thus reduce the accuracy of anomaly detection.

To address the aforementioned challenges, this article proposes a hyperspectral anomaly detection method based on the tensor under prior physical constraints. To completely retain the geometric characteristics and physical rules of the data, we use the tensor to describe it in this method. The proposed method's flowchart is depicted in Fig. 1. The image tensor is decomposed into two parts, the anomaly tensor and the background tensor, which are treated independently. Physically, the background tensor has a wide sample distribution and little spectral change. Based on this, we apply piecewise smoothing constraints on it using linear total variation (TV) norm regularization, which might lessen the introduction of extra noise while restoring the background. Moreover, a low-rank constraint based on image eigenvalues is designed to lessen the influence of anomalies on background learning in its spatial dimension, producing the background model with strong representational capacity. It is discovered that when unfolded along the spectral dimension, the anomalous tensor exhibits sparse characteristics. After a series of iterative optimizations, the anomalies are ultimately identified using the anomaly tensor. The following are the major contributions of this article.

1) The proposed tensor framework organically combines a range of physical prior constraints (such as low rank, sparse, and piecewise smoothing constraints) while completely keeping the data's original geometric

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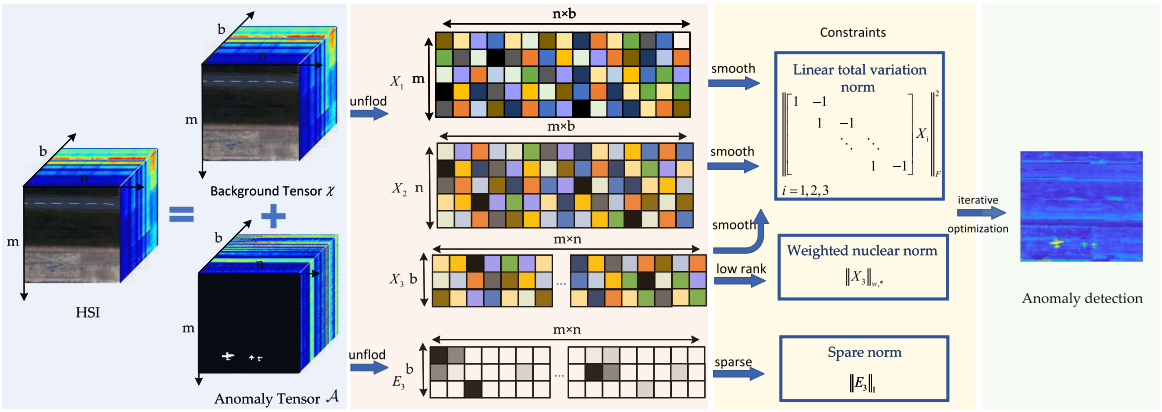


Fig. 1. Flowchart of the proposed method for hyperspectral anomaly detection.

features. As a result, a more expressive background model can be obtained and the practical application ability of the method can be enhanced.

2) A weighted matrix relating to the eigenvalues of image is designed using the characteristics of background and anomaly distribution. Due of their huge coefficients, anomalies are gradually subtracted from the background during iterative optimizations. This enables us to gather trustworthy background data and enhance the precision of anomaly detection.
3) Multiple experiments have been conducted on four datasets, and the results demonstrated that the method is resistant to complicated backgrounds and sensitive to tiny anomalies. The method obviously performs superior to the state-of-the-art methods by evaluating several frequently used metrics.

II. Proposed Framework

A. Notations

This section introduces some of the symbols that will be used later. Matrices are denoted by uppercase letters, e.g., X, vectors are denoted by lowercase letters with a lower corner mark, e.g., x_i, and elements are denoted by lowercase letters or lowercase letters with two (or more) lower corner marks, e.g., a or x_{ij}. Tensors are the generalization of vectors and matrices in higher dimensions, denoted by uppercase cursive letters, such as $\mathcal{X}$. The unfolding of the tensor is represented in uppercase letters with a numerical superscript, such as X_1.

Given an HSI $Y \in \mathbb{R}^{m\times n\times b}$, its 2-D spatial dimensions are $m \times n$ and it has b bands. Denote the (i, j, k)th element of the matrix by x_{ijk}.

B. Method

An HSI is a 3-D data, containing two spatial dimensions and one spectral dimension. Most methods convert it into 2-D data for processing, which will lose the image's original context and structural details in the process. A type of expression known as a tensor, which was first introduced toward the end of the 19th century, preserves geometrical characteristics and natural laws when coordinates are changed. In addition, it has shown to be a useful tool for equally characterizing spatial and spectral data. Here, we use the tensor to describe HSI data.

A given HSI, represented by a tensor $\mathcal{Y} \in \mathbb{R}^{m\times n\times b}$, is a third-order tensor with three modes. In this case, modes 1 and 2 are used to describe the two spatial dimensions, while mode 3 is used to describe the spectral dimension. Correspondingly, we can obtain three matrices by unfolding this tensor in each mode, as follows.

1) *Mode-1 unfolding*

$$Y_1 = [\mathcal{Y}(:,:,1), \mathcal{Y}(:,:,2), \ldots, \mathcal{Y}(:,:,B)], \quad Y_1 \in \mathbb{R}^{m\times(n\times b)}. \tag{1}$$

2) *Mode-2 unfolding*

$$Y_2 = \left[\mathcal{Y}(:,:,1)^T, \mathcal{Y}(:,:,2)^T, \ldots, \mathcal{Y}(:,:,B)^T\right], \quad Y_2 \in \mathbb{R}^{n\times(m\times b)}. \tag{2}$$

3) *Mode-3 unfolding*

$$Y_3 = \left[\mathcal{Y}(:,1,:)^T, \mathcal{Y}(:,2,:)^T, \ldots, \mathcal{Y}(:,W,:)^T\right], \quad Y_3 \in \mathbb{R}^{b\times(m\times n)}. \tag{3}$$

The image tensor $\mathcal{Y}$ is made up of two parts: the background tensor $\mathcal{X}$ and the anomalous tensor $\mathcal{A}$, as shown by the equation $\mathcal{Y} = \mathcal{X} + \mathcal{A}$. The samples in the background portion of the HSI are widely dispersed, and the positions of identical ground items are similar geographically. As a result, its spectral curve does not dramatically alter, indicating that the background portion is smooth in two dimensions. In other words, the matrices X_1 and X_2 that the background tensor unfolds along two spatial dimensions satisfy the feature of local smoothness. We observed that the spectral resolution of the HSI is relatively high, allowing each sample to extract a fairly smooth spectral curve. This indicates that the local smoothness quality also exists in the matrix X_3 that the background tensor unfolds along the spectral dimension. Accordingly, the background tensor is piecewise smooth in general. TV norm regularization is widely used for smoothing constraints in hyperspectral processing tasks [48], [49], [50], [51] such as image denoising, hyperspectral unmixing, HSI restoration, and feature extraction. Here, we use it to constrain the background tensor, namely,

$$\min_{X} \ \|D_H X_1\|_F + \|D_W X_2\|_F + \|D_B X_3\|_F \quad \text{s.t. } \mathcal{Y} = \mathcal{X} + \mathcal{A} \tag{4}$$

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where

$$D_H = \begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & \ddots & \ddots & \\ & & & 1 & -1 \end{bmatrix} \in \mathbb{R}^{(m-1)\times m}$$

$$D_W = \begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & \ddots & \ddots & \\ & & & 1 & -1 \end{bmatrix} \in \mathbb{R}^{(n-1)\times n}$$

$$D_B = \begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & \ddots & \ddots & \\ & & & 1 & -1 \end{bmatrix} \in \mathbb{R}^{(b-1)\times b}.$$

Minimizing the TV regularization constraint is a convex function. However, the discontinuity of its derivatives makes it difficult to solve. This issue is effectively resolved by the linear TV norm, which substitutes the least-square deviation for the smallest absolute value deviation of neighboring pixels. The objective function is rewritten as follows:

$$\begin{aligned} &\min_{\mathcal{X}} \|D_H X_1\|_F^2 + \|D_W X_2\|_F^2 + \|D_B X_3\|_F^2 \\ &\text{s.t. } \mathcal{Y} = \mathcal{X} + \mathcal{A}. \end{aligned} \tag{5}$$

The model is more susceptible to outliers because of the square of the error. The model magnifies the mistake when the deviation exceeds 1. We are able to exclude anomaly samples from the background tensor by iterative solution. The linear TV norm is better suited and simpler to resolve as a result.

The sample of backgrounds has a finite variety of features and each feature recurs over a large range. As a result, the matrix X_3 has a low-rank feature that can be constrained by the matrix's rank. However, the matrix rank minimization problem is a non-deterministic Polynomial (NP)-hard problem with no exact solution. As its best convex approximation, the nuclear norm is used in the following:

$$\begin{aligned} &\min_{\mathcal{X}} \frac{\tau}{2}\left(\|D_H X_1\|_F^2 + \|D_W X_2\|_F^2 + \|D_B X_3\|_F^2\right) + \alpha\|X_3\|_* \\ &\text{s.t. } \mathcal{Y} = \mathcal{X} + \mathcal{A} \end{aligned} \tag{6}$$

where $\|X_3\|_* = \sum_i \sigma_i(X_3)$ represents the sum of the matrix eigenvalues, and τ and α are the regularization parameters. The definition of the nuclear norm makes clear how equally the matrix's components are handled. But every element of the eigenvalues has a distinct physical significance [52]. With the help of this previous information, we intend to further lessen the impact of anomalies on background modeling. The background samples are evenly scattered throughout the HSIs, and their rate of distribution is significantly higher than that of the anomalous samples. Accordingly, the eigenvalues of the background samples are also significantly higher than those of the anomalous samples. On the basis of this, we design a weight matrix to improve the initial nuclear norm. We chose the weight to be the reciprocal of eigenvalues of the matrix X_3, that is, $w = 1/(\sigma_i(X_3) + \text{eps})$, where eps is a tiny positive value to prevent the denominator from being 0. The objective function is rewritten as follows:

$$\begin{aligned} &\min_{\mathcal{X}} \frac{\tau}{2}\left(\|D_H X_1\|_F^2 + \|D_W X_2\|_F^2 + \|D_B X_3\|_F^2\right) + \alpha\|X_3\|_{w,*} \\ &\text{s.t. } \mathcal{Y} = \mathcal{X} + \mathcal{A} \end{aligned} \tag{7}$$

where $\|X_3\|_{w,*} = \sum_i w_i\sigma_i(X_3)$. When a sample is the anomaly, the value of its associated eigenvalue is tiny and its related weighted coefficient is big. The anomaly rapidly fades away from the background throughout the continuous iterative optimization of the objective function because of its huge coefficient.

The anomalous portion of the HSIs has a small distribution area and a low likelihood of occurrence. When the image is unfolding along the spectral dimension, it can be seen as an embedded sparse component. We use l_1 regularization to constrain it. The objective function is finally as follows:

$$\begin{aligned} &\arg\min_{\mathcal{X},\mathcal{A}} \frac{\tau}{2}\left(\|D_H X_1\|_F^2 + \|D_W X_2\|_F^2 + \|D_B X_3\|_F^2\right) \\ &\qquad +\alpha\|X_3\|_{w,*} + \beta\|A_3\|_1 \\ &\text{s.t. } \mathcal{Y} = \mathcal{X} + \mathcal{A} \\ &\qquad X_1 = \text{unflod}_1(\mathcal{X}) \\ &\qquad X_2 = \text{unflod}_2(\mathcal{X}) \\ &\qquad X_3 = \text{unflod}_3(\mathcal{X}) \\ &\qquad A_3 = \text{unflod}_3(\mathcal{A}) \end{aligned} \tag{8}$$

where β is the regularization parameter.

C. Optimization

In this section, we demonstrate how we solve the objective function using the alternating direction method of multipliers (ADMM) [53]. We can break the original optimization problem down into a number of manageable subproblems for iterative solution using this optimization method.

The augmented Lagrangian function of (8) can be expressed as

$$\begin{aligned} &\mathcal{L}(\mathcal{X}, \mathcal{A}, V_1, V_2, V_3, V_4, P_1, P_2, P_3, P_4, P_5) \\ &= \frac{\tau}{2}\left(\|D_H V_1\|_F^2 + \|D_W V_2\|_F^2 + \|D_B V_3\|_F^2\right) \\ &\qquad +\alpha\|V_4\|_{w,*} + \beta\|A_3\|_1 + \frac{\mu}{2}\sum_{i=1}^{3}\|X_i - V_i - P_i\|_F^2 \\ &\qquad + \frac{\mu}{2}\|X_3 - V_4 - P_4\|_F^2 + \frac{\mu}{2}\|Y_3 - X_3 - A_3 - P_5\|_F^2 \\ &\text{s.t. } V_1 = X_1 \\ &\qquad V_2 = X_2 \\ &\qquad V_3 = X_3 \\ &\qquad V_4 = X_3 \end{aligned} \tag{9}$$

in which V_1, V_2, V_3, V_4 are the intermediate variables, P_1, P_2, P_3, P_4, P_5 are the Lagrangian multipliers, and μ is a Lagrangian penalty parameter. By constantly updating one variable while fixing other variables, we can get the solution of (9). The procedure for updating these variables is described in detail in the sections that follow.

1) *Update V_1:* Problem (9) can be expressed as follows for V_1:

$$\mathcal{L}(V_1) = \frac{\tau}{2}\|D_H V_1\|_F^2 + \frac{\mu}{2}\|X_1 - V_1 - P_1\|_F^2. \tag{10}$$

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We can get the update rule of V_1 by taking the derivative of (10) and setting it to zero

$$V_1 = \mu(X_1 - P_1)\left(\tau D_H^T D_H + \mu I_m\right) \tag{11}$$

where $I_m \in \mathbb{R}^{m\times m}$ is the identity matrix.

2) *Update V_2:* Similar to V_1, the update rule for V_2 is as follows:

$$V_2 = \mu(X_2 - P_2)\left(\tau D_W^T D_W + \mu I_n\right) \tag{12}$$

where $I_n \in \mathbb{R}^{n\times n}$ is the identity matrix.

3) *Update V_3:* Similar to V_1, the update rule for V_3 is as follows:

$$V_3 = \mu(X_3 - P_3)\left(\tau D_B^T D_B + \mu I_b\right) \tag{13}$$

where $I_b \in \mathbb{R}^{b\times b}$ is the identity matrix.

4) *Update V_4:* Problem (9) can be expressed as follows for V_4:

$$\mathscr{L}(V_4) = \alpha\|V_4\|_{w,*} + \frac{\mu}{2}\|X_3 - V_4 - P_4\|_F^2. \tag{14}$$

It can be solved by the weighted singular value threshold function [54], and its update rule is

$$V_4 = S_{w,\frac{\alpha}{\mu}}(X_3 - P_4) = U\left(\Sigma - \tfrac{\alpha}{\mu}\mathrm{diag}(w)\right)_+ V^T \tag{15}$$

where $U\Sigma V^T$ is the singular value decomposition of $(X_3 - P_4)$, and $(\cdot)_+$ is to ensure that the data are nonnegative.

5) *Update X:* Similar to V_1, the update rules for X_1, X_2, and X_3 are as follows:

$$\begin{aligned} X_1 &= V_1 + P_1 \\ X_2 &= V_2 + P_2 \\ X_3 &= \frac{1}{3}(V_3 + P_3 + V_4 + P_4 + Y_3 - S_3 - P_5). \end{aligned} \tag{16}$$

6) *Update A_3:* Problem (9) can be expressed as follows for A:

$$\mathscr{L}(A_3) = \beta\|X_3\|_1 + \frac{\mu}{2}\|Y_3 - X_3 - A_3 - P_5\|_F^2. \tag{17}$$

The updating rules for A_3 are as follows when using the soft threshold function [55]:

$$A_3 = \mathrm{soft}\left(Y_3 - X_3 - P_5, \tfrac{\beta}{\mu}\right) \tag{18}$$

where $\mathrm{soft}(u, v) \equiv \mathrm{sign}(u)\max\{|u| - v, 0\}$, and $\mathrm{sign}(\cdot)$ is the sign function.

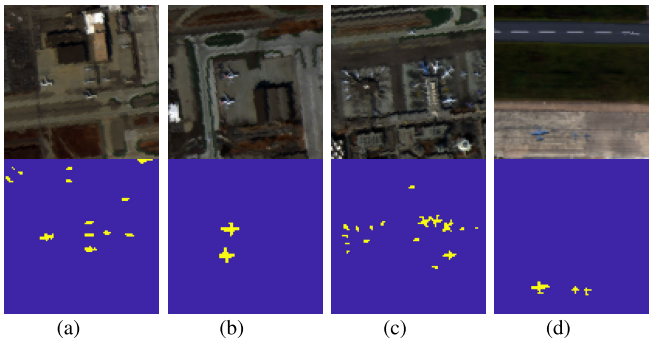
7) *Update the Lagrange multipliers P_1–P_5*

$$\begin{aligned} P_1 &= P_1 - (X_1 - V_1) \\ P_2 &= P_2 - (X_2 - V_2) \\ P_3 &= P_3 - (X_3 - V_3) \\ P_4 &= P_4 - (X_3 - V_4) \\ P_5 &= P_5 - (Y_3 - X_3 - A_3). \end{aligned} \tag{19}$$

The proposed variables are iteratively updated to converge in accordance with the aforementioned update rules. The anomaly detection map S can be obtained from anomaly tensors

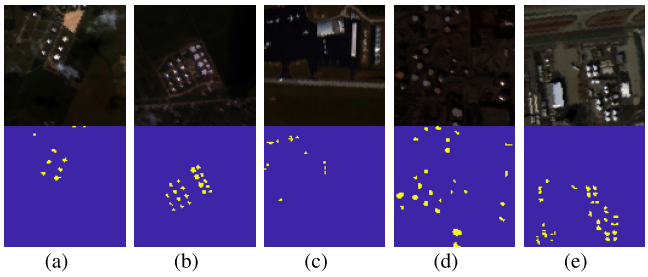
$$S_{i,j} = \left\|\sum_k \mathcal{A}(i, j, k)\right\|_2 \tag{20}$$

where $\mathcal{A} = \mathrm{flod}_3(A_3)$.



(a) (b) (c) (d)

Fig. 2. False color images and visualization of ground-truth masks of the Airport dataset. (a) Airport-1. (b) Airport-2. (c) Airport-3. (d) Airport-4.



(a) (b) (c) (d) (e)

Fig. 3. False color images and visualization of ground-truth masks of the Urban dataset. (a) Urban-1. (b) Urban-2. (c) Urban-3. (d) Urban-4. (e) Urban-5.

III. Experimental Result

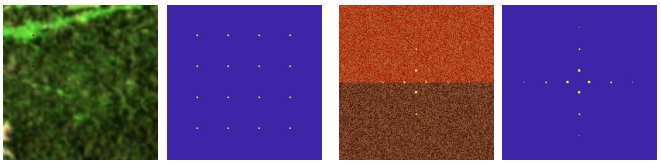
In this section, we evaluate the performance of the proposed method by comparing it with other methods. The experiments are conducted on two genuine datasets and two simulated datasets. The receiver operating characteristic curve (ROC) and the area under the curve (AUC) are the assessment metrics used. The detection probability (PD) and false-positive rate (FAR) of each potential threshold are plotted using the ROC. The AUC is a digital representation of the method's detection performance. Simultaneously, a separability map is introduced to further compare the method's performance. The detection results represent the extent of separation between the background and the anomalies.

A. Dataset Description

The first dataset is the Airport dataset, which was gathered using the airborne visible infrared imaging spectrometer (AVIRIS), as shown in Fig. 2. There are four images overall, each measuring 100 × 100 . The first three images were shot in Los Angeles, and the fourth one was captured in Gulf-port. After removing several bands affected by water absorption and low signal-to-noise ratio, only 205, 205, 205, and 191 bands were retained for each of the four images.

The second dataset is the Urban dataset, which was also collected using AVIRIS, as shown in Fig. 3. There are five images overall, each measuring 100 by 100 pixels. The first two images were shot along the Texas coast, the third one was taken in Gainesville, and the rest were all taken in Los Angeles. The 204, 207, 191, 205, and 205 bands were kept for each of the five images after a number of low signal-to-noise ratio and water absorption bands were eliminated.

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(a) (b)

Fig. 4. False color images and visualization of ground-truth masks of the Simulation datasets. (a) Simulation-1. (b) Simulation-2.

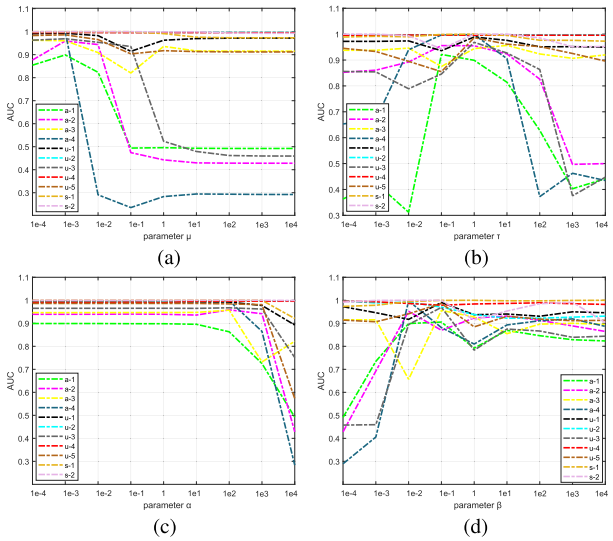
The third dataset is Simulation-1, which has 126 bands and a size of 100 × 100, as shown in Fig. 4(a). In this image, anomalous targets are inserted into four rows and four columns with varying percentages of anomalous spectra over a background of actual grass images. The synthetic formula for anomalous targets is $A = B \times (1 - p) + T \times p$, where A is the spectrum of anomalous target, B is the background average spectrum, T is the anomalous spectrum, and p is the background mixed ratio. The values of p for the four columns of anomalous targets are, in order, 0.8, 0.5, 0.5, and 0.3.

The fourth dataset, Simulation-2, has 105 bands and an image size of 200 by 200, as shown in Fig. 4(b). To evaluate the effectiveness of various methods, this dataset simulates complicated background distributions using the spectra of several materials. The first half of the background sample is made up of spectra of long grass and turf grass combined in random proportions, while the second half is made up of spectra of black brush leaves and sage brush mixed in random proportions. The three sizes of the anomalous targets are 1 × 1, 2 × 2, and 3 × 3, which correspond to the pure spectra of tumbleweed, long grass, and sagebrush, respectively.

B. Comparison Methods

The proposed method is compared with 14 different comparison methods in the dataset above. Among the comparison methods, RX [21], FrFT [24], anomaly detection method via integration of feature extraction and background purification (FEBPAD) [25], and 2-step method based on the principle of the generalized likelihood ratio test (2S-GLRT) [23] are the statistical-model-based methods. In addition, the spectral-spatial based local summation anomaly detection method (LSAD) [28] and CRDBPSW [29] are modifications to the basic CRD method, and anomaly detection based on low-rank and sparse representation (LRASR) [30], low-rank and sparse matrix decomposition based Mahalanobis distance method (LSMAD) [33], method based on graph and TV Regularized low-rank representation (GTVLRR) [34], and method based on visual attention model and background subtraction with adaptive weight (VABS) [11] are the representation-model-based methods. The tensor-based method based on tensor principal component analysis (TPCA) [45] and prior-based tensor approximation method (PTA) [37] are used as the comparison methods. Specifically, we additionally compared two deep-learning-based methods, Auto-AD [17] and LREN [18], on genuine datasets.

The parameters for these methods are configured using the recommended optimal values as provided in the corresponding original literature. For the FEBPAD method, we set the fractional transform order p to 0.7, the maximum rank of the background L to 3, and the sparse cardinality of the dictionary matrix k to 0.7. For the LSAD method, the inner window size w_{in} is 3, the outer window size w_{out} is 5, and the parameter λ is 0.01. The CRDBPSW method's inner and outer window sizes are equal with LSAD, while the other parameter λ is 10^{-6}. The sparse cardinality k in the LSMAD method is 0.005, while the background matrix's maximum rank r is 2. In the 2S-GLRT method, its outer window size w_{out} ranges from 5 to 25, and the inner window size w_{in} ranges from 3 to 15. In the GTVLRR method, the number of nearest neighbors k is set to 10, the scalar parameter σ is set to 1, and the tradeoff parameters λ, β, and γ are set to 0.5, 0.2, and 0.05, respectively. In the VABS method, the number of layers of the pyramid is set to 5, and the proportions c and ε are chosen from $\{1, 2, 3\}$ and $\{1, 2\}$, respectively. For the PTA method, the value of the truncated low-rank coefficient r is determined from $[0, 20]$, the value of the regularization parameter μ is determined from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10^{1}, 10^{2}, 10^{3}, 10^{4}\}$, and the other parameters α, δ, and β are 1, 1, and 0.01, respectively.



(a) (b) (c) (d)

Fig. 5. Four parameters' values have a different effect on how the proposed method's AUC varies in different datasets. (a) μ. (b) τ. (c) α. (d) β.

C. Parameter Tuning

The proposed method contains a total of four parameters: the Lagrange penalty parameter μ and the regularization parameters τ, α, and β. The range of values for each parameter is $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10^{1}, 10^{2}, 10^{3}, 10^{4}\}$. After altering any parameter, the other three will be fixed. The value of this parameter will then be determined by comparing the change in the AUC value. Fig. 5(a)–(d) displays a curve showing the changes in AUC values for various values of the parameter. We have abbreviated the names of the four images in the Airport dataset as a-i ($i = 1, 2, 3, 4$), the five images in the Urban dataset as u-i ($i = 1, 2, 3, 4, 5$), and the two Simulation datasets as s-i ($i = 1, 2$) in the figure for convenience.

Fig. 5(a) shows that for the Urban-3 dataset, the highest corresponding AUC value occurs when the μ value is 10^{-4},

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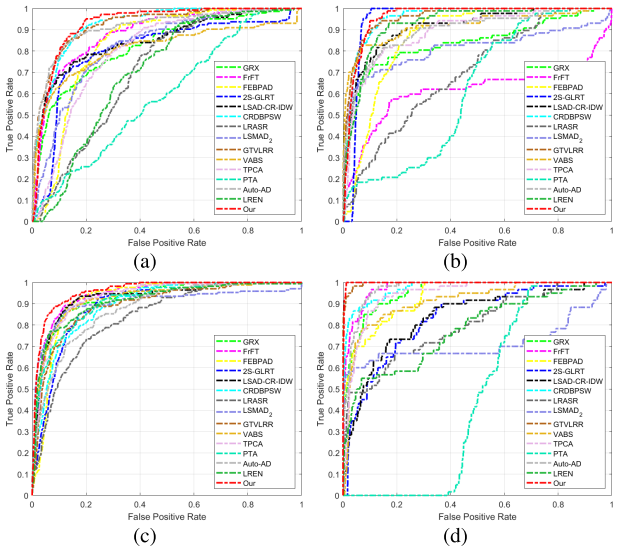


Fig. 6. ROCs of the proposed method and the comparison methods on the Airport dataset. (a) Airport-1. (b) Airport-2. (c) Airport-3. (d) Airport-4.

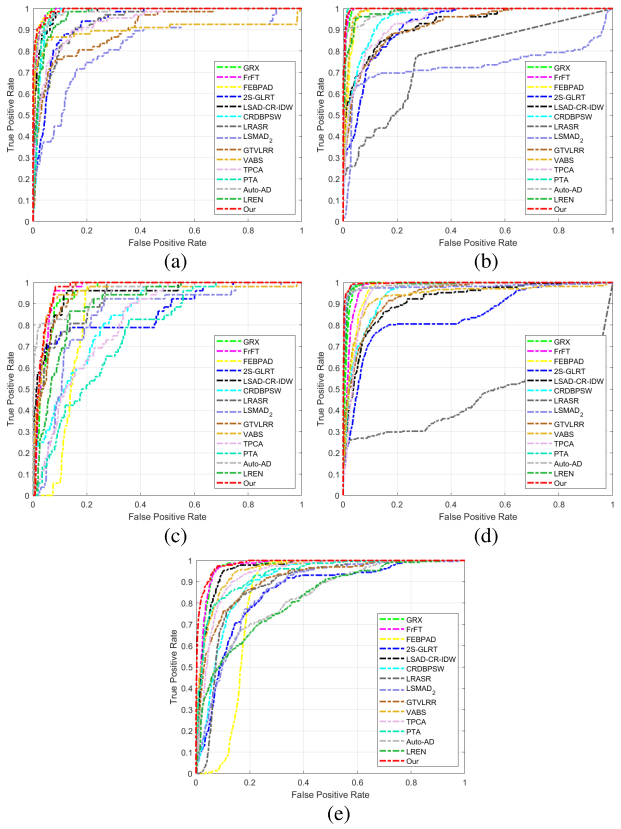


Fig. 7. ROCs of the proposed method and the comparison methods on the Urban dataset. (a) Urban-1. (b) Urban-2. (c) Urban-3. (d) Urban-4. (e) Urban-5.

whereas for the other datasets, the highest corresponding AUC value occurs when the μ value is 10^{-3}. As a result, we set the μ value at 10^{-3}. Fig. 5(b) displays the method's AUC values for several datasets when the parameter τ is changed. As can be observed, the proposed method does not depend on the value of τ for the second, third, and fourth images in the Urban dataset. The method has the maximum AUC value in the Airport-1 and Airport-2 datasets when τ is 0.1, and in the Airport-3 dataset when τ is 10. The method works best in the remaining datasets when τ is set to 1. As a result, we determine τ form $\{0.1, 1\}$. The AUC values for the proposed method in the datasets for various values of α are displayed in Fig. 5(c). The figure demonstrates that the AUC for this method is highest in the Airport-1 dataset at $\alpha = 10$ and highest in the other datasets at $\alpha = 100$. So we simply fix α at 100. Fig. 5(d) displays the influence of various β values on the method's AUC for various datasets. It is obvious that this method is more sensitive to the β value. In the majority of datasets, the AUC value is maximum when $\beta = 10^{-2}$ or $\beta = 10^{-1}$. For best results, β must be specifically tuned for each dataset.

D. Detection Performance

To assess the effectiveness of the proposed method, we conducted experiments on two genuine datasets. The ROCs for the proposed method and the comparison methods are shown in Figs. 6 and 7 respectively. The associated AUC values are shown in Tables I and II, allowing us to quickly compare the detection rates of various methods across various datasets. The numbers in bold represent each method's highest AUC on the current dataset. We discovered that our proposed method had the greatest detection rates on the Airport dataset and two Simulation datasets by comparing both the ROCs and AUC values. In the Urban-2 dataset, the AUC of this method is 99.74%, which is marginally less than the AUC of the PTA method, which is 99.81%. However, our proposed method achieved the highest AUC values in the remaining four images of the Urban dataset, and the average AUC value across all the five images was also the highest.

TABLE I

AUC VALUES (%) OF THE PROPOSED METHOD AND BASELINES USING THE AIRPORT DATASET

Dataset	Airport-1	Airport-2	Airport-3	Airport-4	Average
RX	82.21	84.04	92.88	95.26	88.60
FrFT	88.30	61.66	94.16	97.58	85.43
FEBPAD	83.01	86.71	92.11	93.36	88.80
2S-GLRT	80.60	94.83	89.31	83.00	86.94
LSAD	85.38	91.18	92.76	83.35	88.17
CRDBPSW	92.57	95.65	89.72	97.44	93.85
LRASR	71.02	69.72	85.59	79.60	76.48
LSMAD	82.27	81.00	90.50	71.55	81.33
GTVLRR	92.52	95.66	89.37	99.65	94.30
VABS	81.34	91.88	92.42	91.15	89.20
TPCA	80.23	88.91	92.97	94.32	89.11
PTA	59.22	61.76	91.01	46.36	64.59
Auto-AD	91.33	89.70	87.65	97.04	91.43
LREN	70.56	92.72	91.38	77.74	83.10
Our method	**92.80**	**95.96**	**96.07**	**99.82**	**96.16**

The detection results of several methods on the Airport dataset are visualized in Figs. 8–11. Among them, RX, LSMAD, LRASR, VABS, and Auto-AD are simple to detect for large-sized anomalous targets and simple to miss detection for tiny targets. A high number of backgrounds are incorrectly regarded by the FrFT method as anomalies in the Airport-2 data. Moreover, there is no clear distinction between the background and the anomalies observed in the

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TABLE II
AUC VALUES (%) OF THE PROPOSED METHOD AND BASELINES USING THE URBAN DATASET

Dataset	Urban-1	Urban-2	Urban-3	Urban-4	Urban-5	Average
RX	99.07	99.46	95.13	98.87	96.92	97.89
FrFT	98.92	99.52	96.27	97.97	97.19	97.97
FEBPAD	99.10	98.25	85.47	96.87	83.33	92.60
2S-GLRT	93.72	92.10	86.52	84.80	84.81	88.39
LSAD	98.24	92.74	94.84	91.67	96.11	94.72
CRDBPSW	97.56	95.61	84.92	95.08	90.28	92.69
LRASR	93.78	76.59	92.84	46.68	88.12	79.60
LSMAD	82.25	73.06	84.14	97.92	85.98	84.67
GTVLRR	90.50	92.02	95.20	94.10	91.17	92.60
VABS	90.10	98.69	95.01	93.83	95.52	94.63
TPCA	93.91	94.09	82.25	96.17	93.70	92.02
PTA	97.76	**99.81**	76.59	99.50	93.86	93.50
Auto-AD	98.65	98.76	95.34	98.97	83.14	94.97
LREN	97.34	98.57	90.05	99.18	82.12	93.45
Our method	**99.32**	99.74	**96.92**	**99.67**	**98.62**	**98.85**

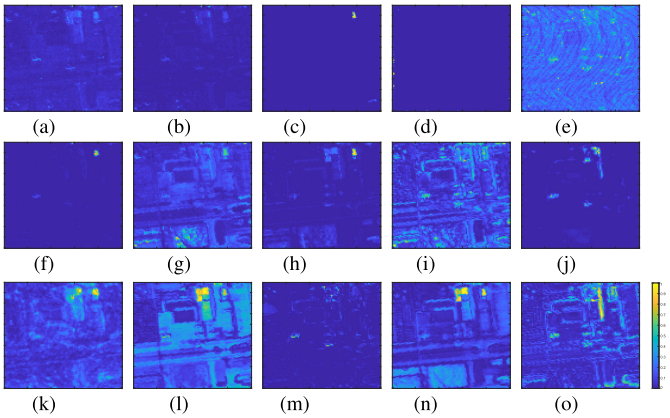


Fig. 8. Detection results for the Airport-1 dataset. (a) RX. (b) FrFT. (c) FEBPAD. (d) 2S-GLRT. (e) LSAD. (f) CRDBPSW. (g) LRASR. (h) LSMAD. (i) GTVLRR. (j) VABS. (k) TPCA. (l) PTA. (m) Auto-AD. (n) LREN. (o) OUR.

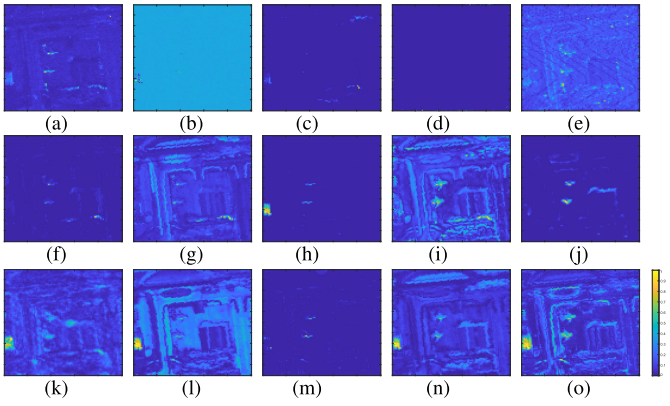


Fig. 9. Detection results for the Airport-2 dataset. (a) RX. (b) FrFT. (c) FEBPAD. (d) 2S-GLRT. (e) LSAD. (f) CRDBPSW. (g) LRASR. (h) LSMAD. (i) GTVLRR. (j) VABS. (k) TPCA. (l) PTA. (m) Auto-AD. (n) LREN. (o) OUR.

detection results. The anomalous likelihood of some pixels is significantly higher than that of other pixels in the detection results of the FEBPAD, 2S-GLRT, and CRDBPSW methods, and therefore, the identified anomalous targets cannot be seen clearly after visualization. Despite the fact that LSAD, GTVLRR, TPCA, PTA, and LREN can all identify all the

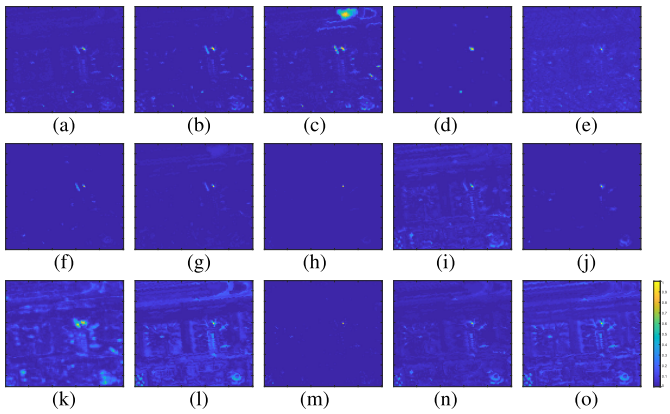


Fig. 10. Detection results for the Airport-3 dataset. (a) RX. (b) FrFT. (c) FEBPAD. (d) 2S-GLRT. (e) LSAD. (f) CRDBPSW. (g) LRASR. (h) LSMAD. (i) GTVLRR. (j) VABS. (k) TPCA. (l) PTA. (m) Auto-AD. (n) LREN. (o) OUR.

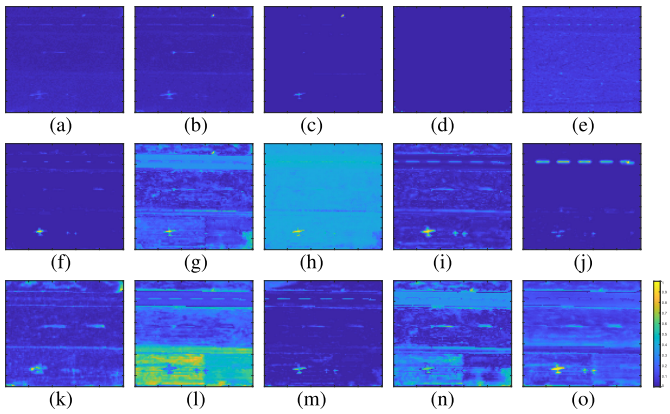


Fig. 11. Detection results for the Airport-4 dataset. (a) RX. (b) FrFT. (c) FEBPAD. (d) 2S-GLRT. (e) LSAD. (f) CRDBPSW. (g) LRASR. (h) LSMAD. (i) GTVLRR. (j) VABS. (k) TPCA. (l) PTA. (m) Auto-AD. (n) LREN. (o) OUR.

anomalous targets, it is simple to include a lot of background information in the results and lessen the salience of anomalies. The proposed method can reliably identify anomalous targets of various sizes, and the distinction between the anomalies and background is clear.

We use the separability maps to further confirm the method's capacity to discriminate between anomalies and backgrounds. A box is used to represent the data in the 20%–80% range, and the horizontal line within the box denotes the data's median value. The data in the intervals of 0%–20% and 80%–100% are shown as dotted lines above and below the box, respectively. To depict anomalies and backgrounds, we use the colors red and blue, respectively. In this case, the method's capacity to identify anomalies improves with increasing distance between the background box and the anomaly box. The proposed method's anomaly box and background box are spread out in every image in the Airport dataset, as shown in Fig. 12.

The detection results of different methods on the Urban dataset are illustrated in Figs. 13–17. LSMAD incorrectly identified a portion of the background in the Urban-1 and Urban-2 data as the anomalous targets, leading to detection maps that were very different from the actual results.

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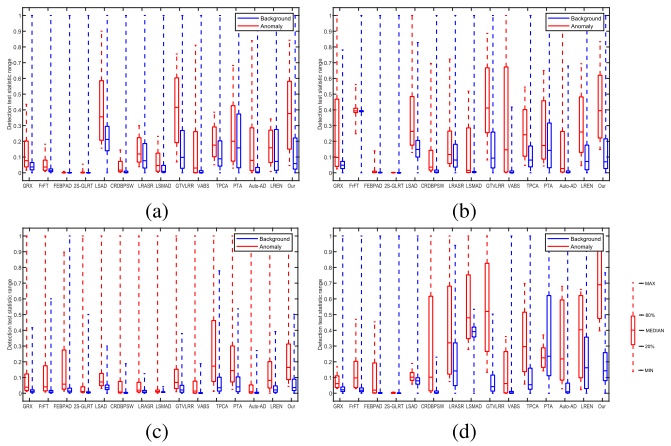


Fig. 12. Separability maps of the proposed method and comparison methods on the Airport dataset. (a) Airport-1. (b) Airport-2. (c) Airport-3. (d) Airport-4.

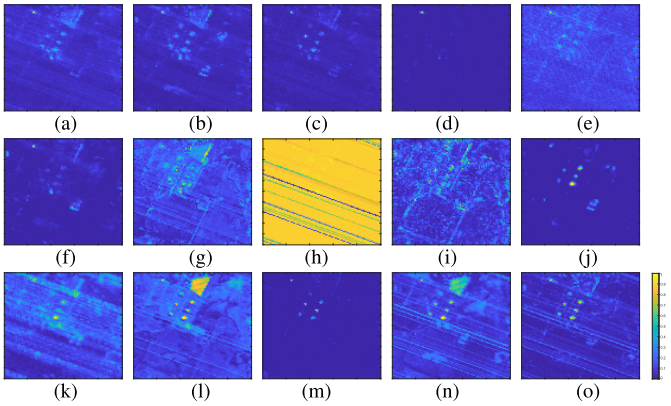


Fig. 13. Detection results for the Urban-1 dataset. (a) RX. (b) FrFT. (c) FEBPAD. (d) 2S-GLRT. (e) LSAD. (f) CRDBPSW. (g) LRASR. (h) LSMAD. (i) GTVLRR. (j) VABS. (k) TPCA. (l) PTA. (m) Auto-AD. (n) LREN. (o) OUR.

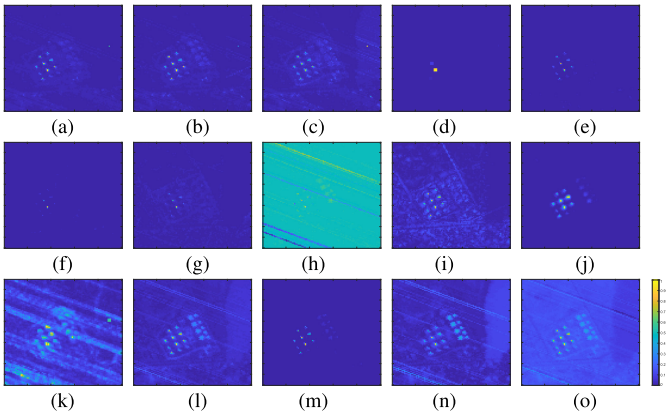


Fig. 14. Detection results for the Urban-2 dataset. (a) RX. (b) FrFT. (c) FEBPAD. (d) 2S-GLRT. (e) LSAD. (f) CRDBPSW. (g) LRASR. (h) LSMAD. (i) GTVLRR. (j) VABS. (k) TPCA. (l) PTA. (m) Auto-AD. (n) LREN. (o) OUR.

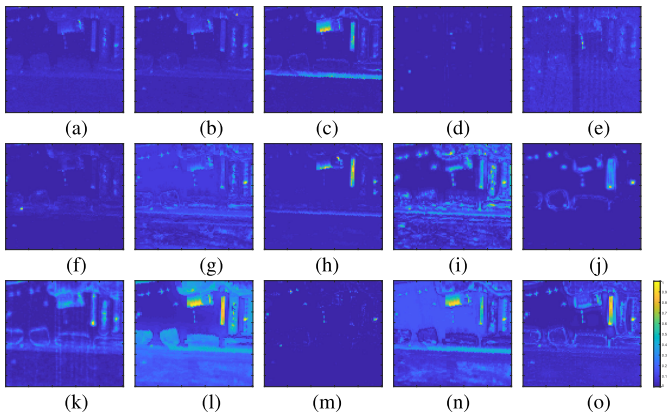


Fig. 15. Detection results for the Urban-3 dataset. (a) RX. (b) FrFT. (c) FEBPAD. (d) 2S-GLRT. (e) LSAD. (f) CRDBPSW. (g) LRASR. (h) LSMAD. (i) GTVLRR. (j) VABS. (k) TPCA. (l) PTA. (m) Auto-AD. (n) LREN. (o) OUR.

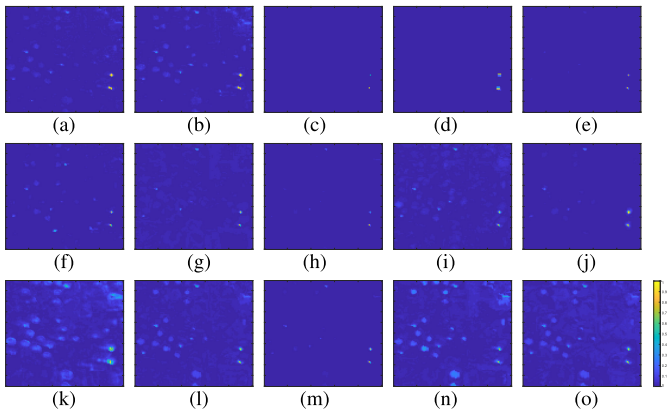


Fig. 16. Detection results for the Urban-4 dataset. (a) RX. (b) FrFT. (c) FEBPAD. (d) 2S-GLRT. (e) LSAD. (f) CRDBPSW. (g) LRASR. (h) LSMAD. (i) GTVLRR. (j) VABS. (k) TPCA. (l) PTA. (m) Auto-AD. (n) LREN. (o) OUR.

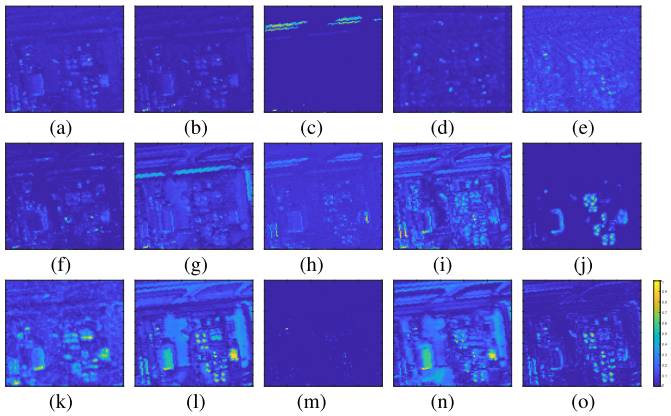


Fig. 17. Detection results for the Urban-5 dataset. (a) RX. (b) FrFT. (c) FEBPAD. (d) 2S-GLRT. (e) LSAD. (f) CRDBPSW. (g) LRASR. (h) LSMAD. (i) GTVLRR. (j) VABS. (k) TPCA. (l) PTA. (m) Auto-AD. (n) LREN. (o) OUR.

The visual detection results of the FEBPAD and 2S-GLRT methods are still not understandable and straightforward, similar to the prior dataset. The detection results of some datasets from LRASR, GTVLRR, TPCA, and PTA contain a significant quantity of background pixels, which lessens the separability of anomalies and backgrounds. All the anomalies can be found using the other methods. Nonetheless, the outlines of anomalous targets are distinct and stand out significantly from the background in the visualization of the proposed method. Fig. 18 further shows that the anomaly box and the

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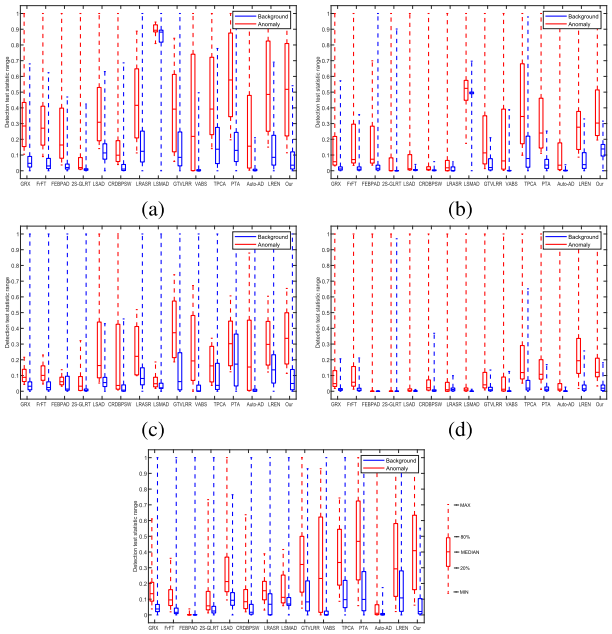


Fig. 18. Separability maps of the proposed method and comparison methods on the Urban dataset. (a) Urban-1. (b) Urban-2. (c) Urban-3. (d) Urban-4. (e) Urban-5.

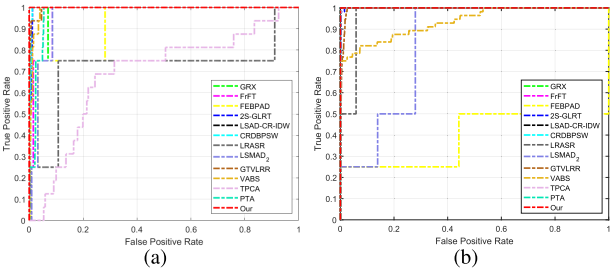


Fig. 19. ROCs of the proposed method and the comparison methods on the Simulation datasets. (a) Simulation-1. (b) Simulation-2.

background box of the proposed method are widely apart, making it simpler to discern between the anomalies and the background.

To further assess the detection capabilities of the traditional methods, we conducted evaluations on two additional augmented simulated datasets. These two datasets were specifically used to compare the sensitivity of different methods to targets with spectra of varying scale of anomalous spectral and to gauge the robustness of these methods in the presence of complex backgrounds. In the relevant experimental results, ROCs are depicted in Fig. 19, while the corresponding AUC values are presented in Table III. The separability maps are illustrated in Fig. 20. Through a comprehensive comparison of AUC values and the distance between the two bounding boxes in the separability maps, our proposed method demonstrates superior performance in detecting anomalies amidst complex backgrounds, exhibiting heightened sensitivity toward anomalous targets.

Furthermore, ablation experiments were conducted on two real-world datasets to assess the impact of image eigenvalue-based weighted matrix and piecewise smoothing constraint

TABLE III

AUC VALUES (%) OF THE PROPOSED METHOD AND TRADITIONAL BASELINES USING SIMULATION DATASETS

Dataset	Simulation-1	Simulation-2	Average
RX	98.12	100	99.06
FrFT	97.04	100	98.52
FEBPAD	91.21	38.96	65.09
2S-GLRT	99.37	99.85	99.61
LSAD	99.99	100	100
CRDBPSW	99.60	100	99.80
LRASR	71.65	97.06	84.36
LSMAD	95.99	82.55	89.27
GTVLRR	99.59	99.49	99.54
VABS	98.96	93.57	96.27
TPCA	68.74	99.96	84.35
PTA	97.52	100	98.76
Our method	**99.99**	**100**	**100**

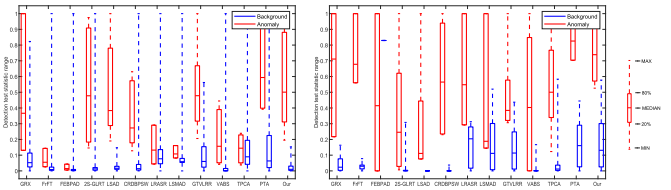


Fig. 20. Separability maps of the proposed method and comparison methods on the Simulation dataset. (a) Simulation-1. (b) Simulation-2.

TABLE IV

PROPOSED METHOD RUNS ABLATION EXPERIMENTS ON TWO DATASETS AND RECORDS THE AUC VALUES (%) IN THE TABLE

Method	method w/o weighted matrix	method w/ two TV norm	method w/ four TV norm	original method
Airport-1	84.30	90.50	49.86	**92.80**
Airport-2	43.33	94.32	54.31	**95.96**
Airport-3	92.02	95.74	87.49	**96.07**
Airport-4	29.59	99.76	30.59	**99.82**
Urban-1	85.92	99.24	96.04	**99.32**
Urban-2	99.73	99.70	**99.80**	99.74
Urban-3	40.17	96.86	69.82	**96.92**
Urban-4	99.59	99.67	52.21	**99.67**
Urban-5	92.35	98.62	92.89	**98.62**

on the spectral dimension for background and anomalies in the proposed method. The corresponding results are summarized in Table IV where "method w/o weighted matrix" refers to the method without the weighted matrix based on image eigenvalues, "method w/ two TV norm" refers to the method without piecewise smoothing constraint on the spectral dimension for background, and "method w/ four TV norm" refers to the method with additional piecewise smoothing constraint on the spectral dimension for anomalies. The results show that except for the Urban-2 dataset, the method w/ four TV norm achieves slightly higher detection accuracy compared with the original method, while the original method outperforms others in the remaining datasets. According to the calculated results, removing the weighted matrix based on image eigenvalues leads to an average 2.62% decrease in AUC, omitting the piecewise smoothing constraint on the spectral dimension for background results in an average 0.06% decrease in AUC, and incorporating the piecewise smoothing

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TABLE V
EXECUTION TIME (SECONDS) OF VARIOUS METHODS ON TWO GENUINE DATASETS AND TWO SIMULATED DATASETS

Method	RX	FrFt	FEBPAD	2S-GLRT	LSAD	CRDBPSW	LRASR	LSMAD	GTVLRR	VABS	TPCA	PTA	Auto-AD	LREN	Our
Airport-1	0.09	12.48	2.17	55.16	13.78	186.69	17.57	6.30	101.35	2.22	17.06	18.93	14.57	182.43	47.62
Airport-2	0.09	12.27	2.30	55.16	13.73	183.43	17.65	6.30	96.12	2.15	16.53	19.05	10.21	179.45	56.83
Airport-3	0.07	12.40	2.23	55.36	13.79	181.48	15.86	6.52	45.39	2.15	16.39	19.18	13.51	183.61	57.47
Airport-4	0.07	11.69	1.89	48.10	13.52	183.97	17.07	5.79	43.70	2.14	16.05	19.98	7.17	166.5	53.66
Urban-1	0.06	12.05	2.06	54.59	13.59	176.93	17.76	6.03	43.96	2.10	15.99	19.39	12.34	221.06	52.86
Urban-2	0.07	12.69	2.06	56.01	13.77	184.10	16.38	6.26	39.88	2.13	16.97	18.77	7.6	231.76	50.67
Urban-3	0.06	11.59	2.06	47.80	13.39	178.10	16.88	5.82	48.94	2.04	16.18	17.31	15.19	173.68	53.10
Urban-4	0.07	12.23	2.05	56.19	13.66	180.54	12.86	6.38	37.52	2.21	16.97	19.09	8.2	228.51	53.10
Urban-5	0.07	12.50	2.08	55.15	13.94	180.97	16.53	6.11	45.88	2.11	16.85	19.24	16.33	203.05	51.77

constraint on the spectral dimension for anomalies yields an average 3.04% decrease in AUC.

The execution times of several methods on two genuine datasets and two simulated datasets are shown in Table V. In all the datasets, the RX method executes in the smallest amount of time, while the CRDBPSW and LREN methods take relatively longer. Due to the requirement for iterative optimization, the proposed method is not time-efficient.

IV. CONCLUSION

In this article, a tensor framework for hyperspectral anomaly detection is proposed. To as accurately separate background and anomalies as feasible, this method can naturally incorporate various physical prior limitations while maintaining the structural information of the data. We divide image tensors into an anomaly tensor and a background tensor. This proposed method uses a linear TV constraint that is based on segmented smooth features in both the spatial and spectral domains of the background. This can improve the immunity to complicated backgrounds by lowering the noise that is produced when the backdrop is restored. According to the background and anomaly distribution criteria, a weighted matrix related to image eigenvalues is designed. Due to the fact that the anomaly coefficient is significantly bigger than the background, it is gradually reduced during the iterative process, enabling the approach to get correct background data and improve sensitivity to tiny anomalies. The l_1-norm is then used to separate sparse anomalies from the background. Two genuine datasets and two simulated datasets were used to evaluate it. By comparing multiple widely used metrics, the proposed method has superior performance. Also, it is capable of accurately separating anomalies from backgrounds.

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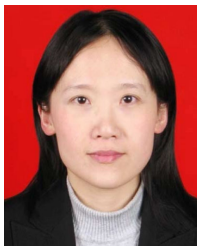
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